higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FUNDAMENTALS
NQF LEVEL 2

19 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1
- An employee or employees could sustain injuries.
- Medical expenses to pay for medical treatment.
- Cost of damage to equipment or materials.
- Increased insurance cost.
- Cost of hiring and training replacement employees.
- Loss of work time. (6 × 2) (12)

1.2
1.2.1 Skills Development Act
1.2.2 The Employment Equity Act
1.2.3 The Basic Conditions of Employment Act
1.2.4 Employment Equity Act/Labour Relations Act
(4 × 1) (4)

1.3
1.3.1 Mechanical industry
1.3.2 Electrical industry
1.3.3 Electrical industry
1.3.4 Automotive industry
1.3.5 Automotive industry
1.3.6 Mechanical industry
(6 × 1) (6)

1.4
- To support the rights and responsibilities of employers and employees with regard to fair labour practices
- To make sure that disciplinary action is carried out in a fair and correct way
- To promote peace in the workplace by applying the Constitution and Bill of Rights
- To encourage employees to take part in the decision-making process
- To allow workers to join a trade union
- To allow employers to belong to an employer's organisation
- To put bargaining councils in place to negotiate with employers about issues such as
- wages of employees (8)

1.5
1.5.1 True
1.5.2 True
1.5.3 False
1.5.4 False
1.5.5 False
1.5.6 True
1.5.7 True
1.5.8 True
1.5.9 False
1.5.10 True
(10 × 1) (10)
[40]

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QUESTION 2

2.1 They can be removed by mechanical safeguarding✓and better machine design and construction.✓ Safe working habits can also contribute towards eliminating injuries.✓ (3)

2.2 2.2.1
- Working at unsafe speed
- Fooling, teasing or abusive co-workers
- Adjusting or working on moving equipment
- Failure to use safety equipment
- Taking chances

2.2.2
- Lack of machine guards
- Unsafe construction
- Overcrowding in a workshop
- Defective working conditions
- Disorder and haphazard planning

(5 × 2) (10)

2.3
- Flameproof gloves
- Leather gloves
- Rubber gloves
- Insulating gloves (4)

2.4
- Unsafe conditions
- Lack of knowledge
- Physical or mental defect
- Improper attitude
- Ignorance. (4)

2.5
- Strong, sound floor on which to build stacks.
- No stack should be more than three times as high as it is wide.
- Stacks should be bonded and interlocked.
- Choose the site for the stack with care and avoid projection.
- No stack should obstruct firefighting equipment.
- Stack in danger of collapse should be broken down immediately.
- Climbing onto stack without the aid of ladders should be prohibited.

(Any 5 × 1) (5)

2.6
- Red
- Yellow
- Green
- Light green (4)

[30]

QUESTION 3

3.1 3.1.1 It will remove the heat.
3.1.2 It will cut off oxygen.
3.1.3 Fire will extinguish itself.
(3 × 1) (3)

3.2 3.2.1 Ordinary combustibles
3.2.2 Flammable
3.2.3 Electrical field
3.2.4 Combustible metals
(4 × 1) (4)

3.3
- The soda acid type discharging water
- The foam type discharging foam
- The dry chemical type discharging powder
- The CO_2 type discharging gas (Any 3 × 1) (3)

[10]

QUESTION 4

4.1
- State your name clearly.
- Give the telephone number you are calling from, in case the connection is cut off.
- Give the exact location or landmarks in the area.
- State the type of injuries or conditions.
- Answer all questions and do not disconnect until instructed to do so.

(Any 3 × 1) (3)

4.2 Fainting is most commonly caused by a temporary shortage of blood✓ and oxygen to the brain.✓ (2)

4.3
- Bad news or a bad fright
- Long periods of standing, especially in hot weather
- Fatigue
- Hunger
- Lack of fresh air (Any 2 × 1) (2)

[7]

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QUESTION 5

5.1
- To protect employees against dangers to health in their working environment
- To place individual workers in jobs suitable to their physical abilities
- To encourage workers to protect their health (3)

5.2
- Safety posters are the most clearly visible evidence of accident prevention work.
- Posters properly used can have great value in a safety programme, as they can influence the attitude and behaviour of the workers.
- They should be located near the entrance of a building.
- Posters should be large enough and instructions should be clear.
- Change posters weekly and record the sequence of display.
- It gives workers the opportunity of giving input on safety issues.

(Any 5 × 1) (5)

5.3
5.3.1 True
5.3.2 True
5.3.3 True
5.3.4 False
5.3.5 False

(5 × 1) (5)
[13]

TOTAL: 100